

1. Which component of a computer display system is responsible for storing the digital bitmapped image before it is rendered on screen?

- (A) Frame Buffer / Video RAM
- (B) ROM BIOS
- (C) Main System Memory (RAM)
- (D) Instruction Register

2. How much Video RAM (VRAM) in Megabytes (MB) is required to store a single uncompressed display frame at a resolution of 1024×768 pixels with 24-bit True Color depth?

- (A) 2.25 MB
- (B) 3.00 MB
- (C) 7.68 MB
- (D) 24.0 MB

3. What is the primary operational difference between raster-scan display systems and vector (random-scan) display systems?

- (A) Raster-scan sweeps line-by-line top-to-bottom; Vector draws lines point-to-point
- (B) Raster-scan uses laser light; Vector uses LED backlighting
- (C) Raster-scan does not require a frame buffer
- (D) Vector displays can only show black-and-white text

4. In laser printer technology, what is the correct sequential order of the electrophotographic printing process?

- (A) Charging → Exposing → Developing → Transferring → Fusing
- (B) Exposing → Charging → Developing → Fusing → Transferring
- (C) Developing → Charging → Exposing → Transferring → Fusing
- (D) Charging → Transferring → Exposing → Fusing → Developing

5. Which display panel matrix technology provides the widest viewing angles and most accurate color reproduction in modern monitors?

- (A) TN (Twisted Nematic)
- (B) IPS (In-Plane Switching)
- (C) VA (Vertical Alignment)
- (D) Passive Matrix LCD

6. What is the function of the Fuser Unit inside a laser printer?

- (A) To charge the photoreceptor drum with negative static electricity
- (B) To apply heat and pressure to permanently melt toner onto paper fibers
- (C) To spray microscopic ink droplets directly onto paper
- (D) To sweep unused paper dust out of the paper tray

7. What is the default aspect ratio of a standard 1080p Full High-Definition (1920×1080) monitor?

- (A) 4:3
- (B) 16:9
- (C) 16:10
- (D) 21:9

8. Which printer language acts as a page description standard developed by Adobe to describe complex vector artwork and typography?

- (A) PostScript
- (B) PCL (Printer Command Language)
- (C) GDI (Graphical Device Interface)
- (D) ESC/P

9. In digital audio output systems, what component converts discrete digital PCM sample values back into a continuous analog voltage signal for speakers?

- (A) ADC (Analog-to-Digital Converter)
- (B) DAC (Digital-to-Analog Converter)
- (C) DSP (Digital Signal Processor)
- (D) MIDI Controller

10. Why are Dot-Matrix Printers still commonly used in railway reservation counters and billing counters?

- (A) Ability to print high-resolution photo prints
- (B) Low noise and extremely fast printing speeds
- (C) Capability to print multi-part carbon copy forms using physical impact
- (D) Wireless cloud printing capability

11. The technical parameter Response Time of a gaming monitor refers to:

- (A) The time required for a pixel to change from one color to another (e.g., Gray-to-Gray)
- (B) The lag between pressing a key and CPU processing
- (C) The time taken to boot up the display backlight
- (D) The speed of image data transmission over HDMI cable

12. Which audio interface standard does NOT carry actual digital sound waveforms, but instead transmits musical event messages (such as pitch, velocity, and note-on/off)?

- (A) S/PDIF
- (B) MIDI (Musical Instrument Digital Interface)
- (C) WAV
- (D) MP3

13. What is the primary function of the RAMDAC chip on legacy computer graphics cards?

- (A) Converting digital pixel data in VRAM into analog signals for CRT monitors
- (B) Accelerating 3D geometry calculations
- (C) Compressing video streams for HDMI transmission
- (D) Storing display calibration profiles

14. Which type of plotter holds paper firmly on a flat rectangular surface while a moveable pen gantry draws vector graphics across both X and Y axes?

- (A) Flatbed Plotter
- (B) Drum Plotter
- (C) Inkjet Plotter
- (D) Electrostatic Plotter

15. What is the principal difference between additive RGB color space and subtractive CMYK color space?

- (A) RGB combines light to form white; CMYK combines pigments to absorb light towards black
- (B) RGB is used for paper printing; CMYK is used for monitor screens
- (C) RGB uses 4 primary colors; CMYK uses 3 primary colors
- (D) CMYK cannot produce black color

16. What visual artifact occurs when a monitor displays fast-moving objects while its refresh rate is not synchronized with the GPU rendering frame rate?

- (A) Screen Tearing
- (B) Pixel Burn-in
- (C) Chromatic Aberration
- (D) Motion Dithering

17. Thermal transfer printing differs from direct thermal printing because thermal transfer:

- (A) Uses a heat-sensitive ink ribbon to melt wax/resin onto plain paper
- (B) Requires chemically coated paper that darkens when heated
- (C) Does not require any ribbon or ink consumable
- (D) Can only print on continuous tractor-feed paper

18. Which display interface cable supports multi-stream transport (MST), enabling daisy-chaining of multiple monitors on a single port?

- (A) DisplayPort
- (B) VGA
- (C) DVI-Single Link
- (D) RCA Composite

19. How many distinct shade levels per primary channel can be represented by a display monitor operating at 8-bit color depth per channel (24-bit True Color)?

- (A) 256 levels (B) 512 levels
(C) 1024 levels (D) 65,536 levels

20. Which non-impact printing mechanism uses piezoelectric crystal deformation or thermal heating elements to create ink droplets?

- (A) Inkjet Printer (B) Laser Printer
(C) Daisy Wheel Printer (D) Line Matrix Printer

21. In computer audio output, what is the Nyquist sampling rate requirement to accurately reproduce human audible frequencies (up to 20 kHz)?

- (A) At least twice the highest frequency (≥ 40 kHz) (B) Equal to the highest frequency (20 kHz)
(C) Four times the highest frequency (80 kHz) (D) Half the highest frequency (10 kHz)

22. What is the purpose of a Print Buffer inside a modern high-speed printer?

- (A) To store print data locally so the CPU can resume other computer operations (B) To clean the printhead after every printed page
(C) To convert RGB images into vector coordinates (D) To regulate internal laser temperature

23. Which aspect of an OLED monitor screen allows it to achieve a virtually infinite contrast ratio?

- (A) Individual pixels can turn off completely to display true absolute black (B) It uses an ultra-bright LED backlight panel
(C) It operates at double the standard refresh rate (D) It uses liquid crystal polarization filters

24. Which legacy impact line-printer uses a continuously rotating horizontal metal chain containing character molds to print entire lines at once?

- (A) Chain Printer (B) Thermal Printer
(C) Inkjet Printer (D) Laser Printer

25. In modern high-end projectors, what does the spec rating ANSI Lumens measure?

- (A) Total light output brightness generated by the projector (B) Screen resolution quality
(C) Audio speaker output power in watts (D) Projection lens focus distance

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(B)	11	(A)	16	(A)	21	(A)
2	(A)	7	(B)	12	(B)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(B)	14	(A)	19	(A)	24	(A)
5	(B)	10	(C)	15	(A)	20	(A)	25	(A)